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Magnetization of the Cambrian Lamotte Formation in Missouri

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Summary

The Upper Cambrian Lamotte formation of South-eastern Missouri in Central United States is a white sandstone which, though very weakly magnetized, is amenable to investigation by standard palaeomagnetic techniques. Out of the total of 150 oriented cores, 134 specimens were found to possess measurable intensities of the order of 1×10^{-6} emu cm^{-3}. Though thermal cleaning does not produce reliable results, magnetic cleaning yields three groupings of significant directions of magnetization, two of which are associated with the lower coarse grained part of the formation and one with its fine grained upper layers. The first two are believed to be associated with secondary magnetization, one representing the direction of the Carboniferous geomagnetic field. The third group subdivides into two subgroups based on sampling of a columnar section near the top of the formation and is associated with the original magnetization of depositional origin. The palaeomagnetic poles of these two subgroups are situated at 160° W, 38° S and 155° E, 11·5° S respectively and are thought to represent two distinct phases of Upper Cambrian geomagnetic field.

Mineralogical studies and investigations of the magnetic properties of the sandstone show that the magnetic minerals carrying the magnetization are members of the ilmenite–haematite series of the same average grain size ($\sim 15\mu$) in both types of rock. The magnetization of the coarse grained part is mainly of chemical origin and is carried predominantly by an homogeneous ilmenite–haematite solid solution, while that of the fine grained part is associated with specular haematite and titanohaematite containing ilmenite lamellae. Both types of sandstone, but especially the coarse grained type, contain titanohaematite with microintergrowths, as well as finely divided haematite in a demagnetized state. Heat treatment of the fine grained sandstone for 6 hr at 735 °C changes its magnetic behaviour to that of the coarse grained type. The results suggest also that the stable part of remanence in natural haematite is that of the lattice defect moment with a ferrimagnetic Curie point at 725 °C.

Comparison of the Upper Cambrian and Carboniferous Lamotte palaeomagnetic poles with the corresponding poles for European rocks suggests that North America and Europe have been drifting apart since Upper Palaeozoic. The results indicate that despite Post-Cambrian remagnetization of the Lamotte sandstone a considerable part of the formation has retained its original Cambrian magnetization.

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1. Introduction

Palaeomagnetic investigations of Cambrian rocks of North America have so far yielded inconclusive results. The pole positions are scattered and it is not clear which of them represent the Cambrian magnetic field. The hypothesis of remagnetization of Lower Palaeozoic rocks of Europe and North America in the Upper Palaeozoic proposed by Creer (1968) calls for caution and suggests that a careful study of the origin of magnetization of selected formations must be conducted before making palaeomagnetic deductions.

The directions of magnetization, obtained by Howell & Martinez (1957) from the Upper Cambrian Sawatch sandy dolomite in Colorado, reflect the direction of the Post-Cambrian, possibly Carboniferous geomagnetic field. The same authors have also investigated the Upper Cambrian siltstone from the Point Peak member of the Wilberns formation in the Llano uplift area in Texas. The directions have a streaked distribution, and the assumed pole was calculated from a group of unit vectors having directions farthest from the present field. Collinson & Runcorn (1960) studied the magnetization of the Lodore formation in Utah and of the Deadwood formation in Wyoming. Day & Runcorn (1955) reported on magnetization of the Tapeats sandstone in Arizona which, however, gave scattered directions of magnetization, and therefore, no statistical analysis was made. The directions in the Deadwood formation were not greatly different from those of the Carboniferous rocks in the USA. The directions in the Lodore formation occurred in distinct positive and negative groups, and the pole positions calculated from these directions were considered to be of Cambrian age. The Cambrian Wichita Mountains igneous complex was studied by Yaskawa, Vincenz & Guja, Ku *et al.* (1967), and Spall (1968). The results revealed scattered directions of magnetization, suggesting that the primary magnetization was masked by secondary components acquired during a period much later than the Cambrian. Careful magnetic and thermal cleaning suggested that the secondary magnetization was of Carboniferous age, but in some cases it was possible to uncover the supposedly primary Cambrian component.

The aim of this paper is to study the natural remanent magnetization (NRM) of the Cambrian Lamotte formation with the hope of resolving the uncertainties described above. There is a certain novelty in the proposed study since very little palaeomagnetic work has so far been done using white sediments such as the Lamotte formation. The white sediments are usually weakly magnetized and the origin of their magnetization is not well understood. It is believed that the study of such weakly magnetized sediments may expand the scope of the palaeomagnetic method and will lead to a more complete record of ancient geomagnetic field.

2. Geology

The Lamotte formation crops out only in the St Francois Mountains region of South-eastern Missouri. The formation is a white sandstone and has been shown to be of Upper Cambrian age, approximately 500 My old, primarily because of its unconformable relationship with the Precambrian rocks and its conformable and gradational contact with the overlying Bonne Terre dolomite (Ojakangas 1960). Radioactive dating of the Precambrian igneous rocks of St Francois Mountains has been performed by Adams (1959), Allen *et al.* (1960), and Tilton, Weatherill & Davis (1962). The Rb–Sr and K–Ar analyses gave ages ranging from 1280 to 1450 My and U–Pb analyses from 970 to 1120 My. According to Tilton *et al.* the latter analyses were affected by loss of lead from the zircons and the corrected U–Pb ages actually range from 1400 to 1550 My in general agreement with the Rb–Sr and K–Ar ages. Palaeontological studies of the Lamotte sandstone clearly indicate its Upper Cambrian age. Although the formation contains few fossils, those present, and in particular

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the fossil trails, allow correlation with the Upper Cambrian Potsdam sandstone of New York and Mt Simon sandstone of Wisconsin (Summerson 1951). The best evidence comes from the fauna of the overlying Bonne Terre formation which is unmistakably Upper Cambrian (Ojakangas 1960) and there is no doubt that the Lamotte sandstone is of the same age. Although the Lamotte formation crops out only in Missouri, it is present in the subsurface throughout Central United States. It is generally about 120 metres thick and geological evidence (James 1951) reveals that it was affected by only very minor movements. These took the form of normal faults with a throw of a maximum of 1 metre and very gentle or no folding. There is clear evidence that by the time of the encroachment of the Upper Cambrian sea in the region of the present day outcrops of the Lamotte sandstone, the underlying Precambrian rocks had already been eroded to a generally flat surface. Thus the bedding of the formation is generally horizontal or has very gentle dips except in area near the Precambrian ridges and knobs where the dips attain about 15° (Ojakangas 1960). An important feature of the formation is that it is white and contains little of the red material so widely spread in many Palaeozoic and Mesozoic sediments. It is convenient to divide the formation into two major rock units:

(*a*) Medium and coarse grained arkose sandstone with conglomerate layers in the lower part; and

(*b*) Fine grained orthoquartzite sandstone which is the dominant rock type of the Lamotte formation.

Although there is some brown or reddish coloration in the lower coarse grained part of the formation, its upper fine grained part is white.

Absence of movement and dip, or very gentle dip, uniformity of composition and fresh appearance of the constituent minerals make the Lamotte formation suitable for palaeomagnetic investigations.

3. Sampling and measurement

Fig. 1 shows a sketch map of the area, indicating the positions of the main outcrops of the formation and of the seven sampling localities. A total of 150 oriented cores was obtained in two samplings conducted during two consecutive summer seasons and covering approximately the full stratigraphic section of the formation. Only horizontally bedded parts of the formation were sampled. The coarse grained arkoses were sampled at localities *B*, *D*, *G* and *E*, the conglomerate at locality *H*, and the fine grained quartzite at localities *C* and *F*. Normally there were several sampling sites in each sampling locality, which were designated by consecutive numbers after the identifying code letter (e.g. *F*1, *F*2), while a second number was assigned to each core and a third to each individual cylindrical specimen (e.g. *H*312). The sampling operations were difficult and the number of cores obtained was severely curtailed by the unfavourable condition of the outcrops. Many of the outcrops were too weathered to allow removal of samples by drilling and the weathered outer layers made it impossible to obtain fresh samples by the block sampling method. The best samples were obtained from road cuts, stream beds or river banks, and man-made lakes. The strongly abrasive property of the sandstone resulted in frequent damage to the steel barrels of drill bits and corresponding contamination of rock cores by steel particles. Thus some cores had to be discarded and special precautions were taken to avoid such a contamination. In the quartzite unit drilling was slow and it was not possible to recover more than 25 per cent of the cores drilled. The arkose unit was easier to drill, but the percentage of the recovered cores was even less than 25 percent. Finally, the formation was found to be so weakly magnetized that a number of cores had to be discarded as too weakly magnetized to allow measurement of their magnetization on the magnetometers used. Thus, eventually, natural intensities of 134 specimens were measured comprising 37 specimens obtained during the first sampling and 97

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during the second sampling. Details of the sampling localities are given in the Appendix.

The direction and intensity of NRM in the specimens from the first sampling were measured on an air-turbine spinner magnetometer (Helsley 1967). All other specimens were measured on Princeton Applied Research (PAR) spinner magnetometer, Model SM-1, and an astatic magnetometer was used for measurement of magnetization imposed in the laboratory. On the spinner magnetometers the error of measurement was less than 1×10^{-7} emu cm^{-3} in magnitude of intensity of magnetization and 1° in direction, while on the astatic magnetometer the corresponding error was 10^{-6} emu cm^{-3} and 5° in direction. Alternating field (AF) demagnetization was conducted using apparatus with a three axis tumbler and capable of developing peak fields up to 2800 Oe. Curie temperatures were determined by means of a conventional Curie balance with fields ranging from 600 to 1200 Oe. The measurements were made exclusively on powdered magnetic material extracted from virgin rock. Thermal demagnetization was performed using apparatus similar to that described by Vincenz, Braught & Myers (1965). WL-POL Zeiss microscope was used in reflected light mineralogical investigations and X-ray diffraction studies were made with a Phillips X-ray diffractometer.

4. Palaeomagnetic data

The virgin directions of magnetization of the specimens obtained in the first sampling are shown in Fig. 2(*a*). Clearly there is too great a scatter in direction to allow any palaeomagnetic conclusions. In addition some of the vectors tend to show a preferential distribution in the direction of the present geomagnetic field in the area of sampling. This suggests the presence of a viscous component in some of the specimens and AF demagnetization was applied to remove it. A set of four pilot specimens was used to determine the optimum peak field suitable for achieving a minimum scatter in directions. Unfortunately no such optimum value for all specimens could be determined and to achieve best results each specimen had to be treated individually.

Fig. 3 shows the directions of the first collection after demagnetization using peak

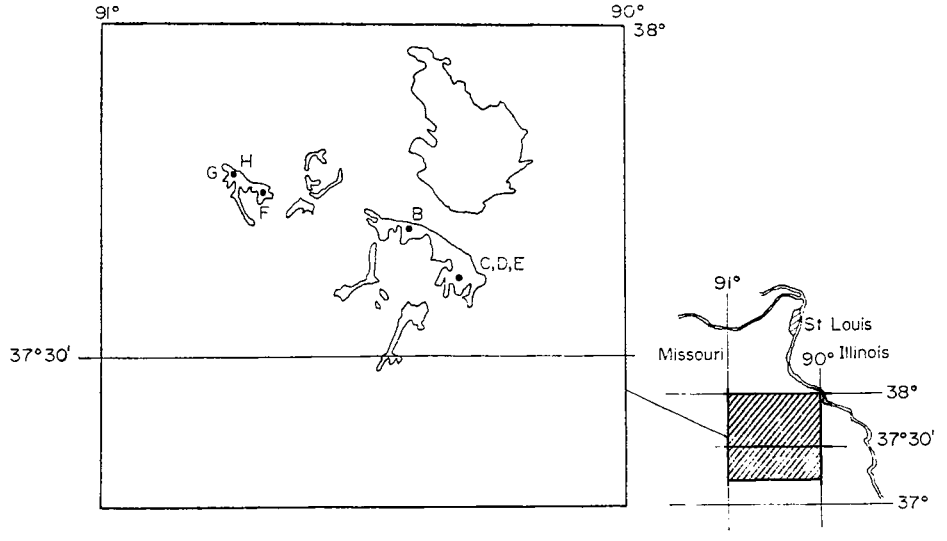


FIG. 1. Geological sketch map of the outcrops of the Lamotte formation. Sampling localities are indicated by full circles.

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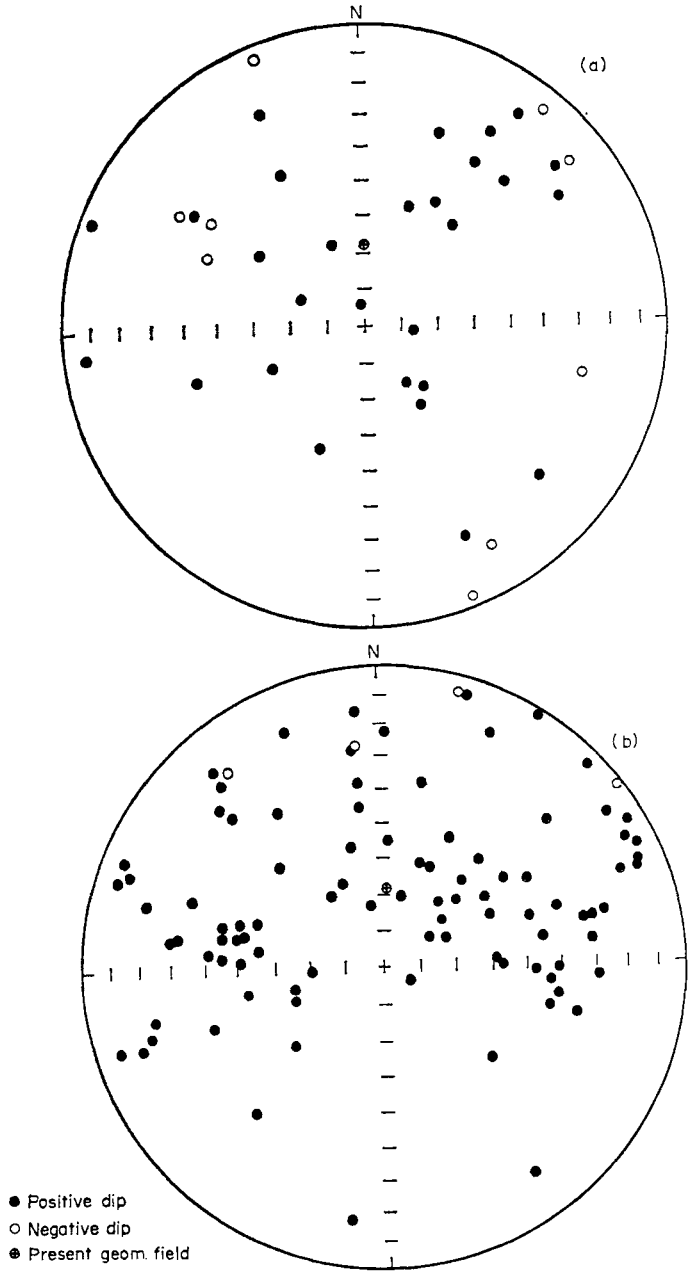


FIG. 2. Equal area projection of virgin NRM directions. (*a*) First collection, (*b*) second collection.

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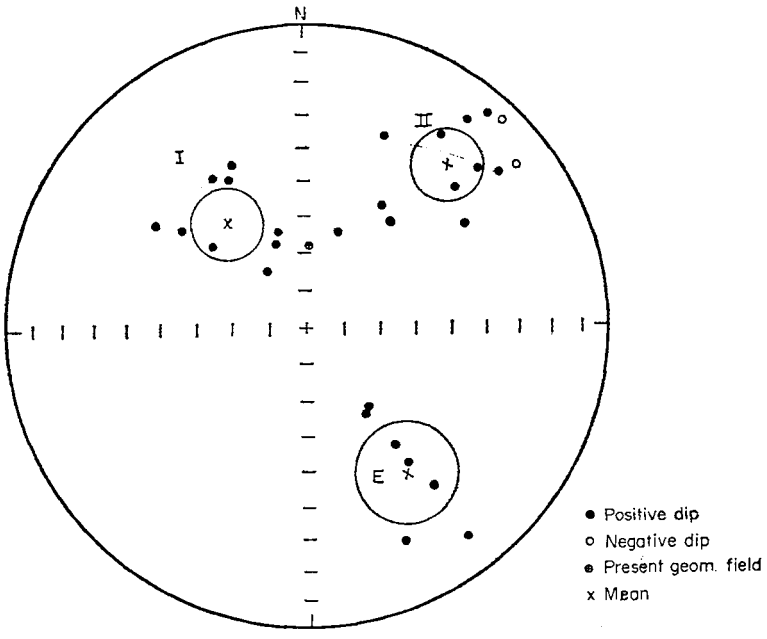


FIG. 3. Equal area projection of NRM directions of the first collection after AF treatment. Group I, demagnetized using peak field of 130 Oe, and group E, demagnetized in 88 Oe, represent the directions of coarse grained arkose sandstone. Group II, demagnetized in peak field of 250 e, represents the fine grained sandstone. Circles around the mean directions represent the 95 per cent cones of confidence (Fisher 1953).

fields of 130 Oe on specimens designated as group I, 88 Oe for the specimens from locality *E*, and 250 Oe for specimens designated as group II. The division into the three groups was made on the basis of the outcome of magnetic 'cleaning' shown in Fig. 3. Group I and locality *E* are associated with the coarse grained lower part of the formation, while group II represents the fine grained sandstone.

An important criterion in the demagnetization procedure was to use maximum peak fields of such a magnitude that allowed the specimens to retain an intensity of magnetization measurable on the magnetometers used. It was found, however, that in more strongly magnetized specimens with virgin directions close to the mean direction obtained after AF demagnetization, even higher peak fields did not produce any significant change in direction.

Fig. 2(*b*) shows the virgin directions obtained for the specimens from the second sampling, excluding the conglomerate samples from locality *H* and Fig. 4 shows these directions after AF demagnetization. Here again the 'clean' directions fall into three distinct groups and the specimens previously designated as groups I and II were demagnetized with peak fields of 250 Oe, while samples from locality E were cleaned using peak fields of 130 Oe. Out of 97 specimens the directions for only 65 were 'cleaned' because the remaining 32 were either too weak to be measured after AF demagnetization or originally very weakly magnetized and unsuitable for AF demagnetization. As before, no significant changes in direction were observed on application of high peak fields to more strongly magnetic specimens which had directions close to the mean direction of each individual group. The results for the second sampling are similar to those for the first, the three mean directions falling in the same sectors of the equal area net. Group II, however, subdivides into two subgroups *a* and *b*. The distinction between *a* and *b* was obtained on the basis of the

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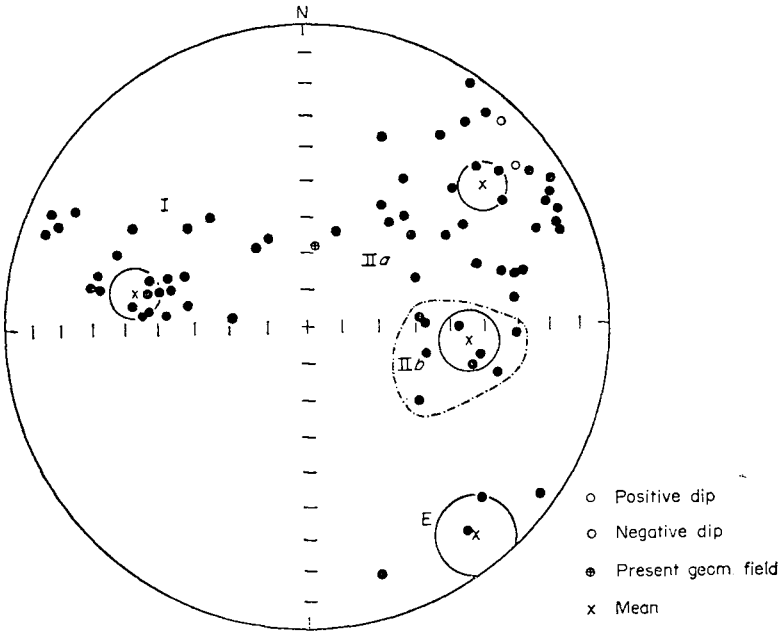


FIG. 4. Equal area projection of NRM directions of the second collection after AF treatment. Specimens of groups I and II cleaned in 250 Oe, and all those from locality E in 130 Oe. Groups II*a* and II*b* represent the lower and upper parts of the quartzite unit respectively.

position of the specimens on different bedding levels of the fine grained sandstone. Specimens of subgroup II*a* were derived from the lower part of the fine grained sandstone, while those of the subgroup II*b* were obtained from the uppermost fine grained beds. This distinction is illustrated in Fig. 5 which shows that the direction of magnetization at site F1 of locality F is is a function of stratigraphic horizon. Clearly the lower part of the columnar section below F11 has a different mean direction than its upper part represented by samples F11 to F17. Since the 95 per cent circles of confidence of the two mean directions, shown in Fig. 4, do not overlap, the two means are statistically different.

An important characteristic of the magnetization of the Lamotte sandstone is its very low intensity. Since the directions of intensities lower than 10^{-7} emu cm^{-3} could not be measured accurately, this imposed a stringent limitation on the measurements of virgin NRM and the limitation became even more severe when magnetic cleaning was conducted. Histograms of the magnitudes of the NRM of the two types of sandstone are given in Fig. 6. Clearly the greater number of specimens (107 out of 144) possess intensities equal to or below 2×10^{-6} emu cm^{-3} and this important aspect of the magnetization of the Lamotte sandstone must be borne in mind throughout this investigation.

It is of interest to note that the intensities of the fine grained rock peak between 1×10^{-6} emu cm^{-3}, but are not distributed symmetrically about the peak (Fig. 6(*a*)). Assuming that the distribution is Poissonian, its variance was computed and the expected Poissonian distribution deduced to fit this assumption. The chi-square goodness of fit test applied to the observed and expected frequencies showed that at the 5 per cent probability level the observed distribution differed significantly from a Poissonian distribution. The observed distribution approximates, in fact, to a logarithmic normal distribution previously noted in specularite bearing sediments by Irving, Molyneux & Runcorn (1966). On the other hand the intensity histograms for

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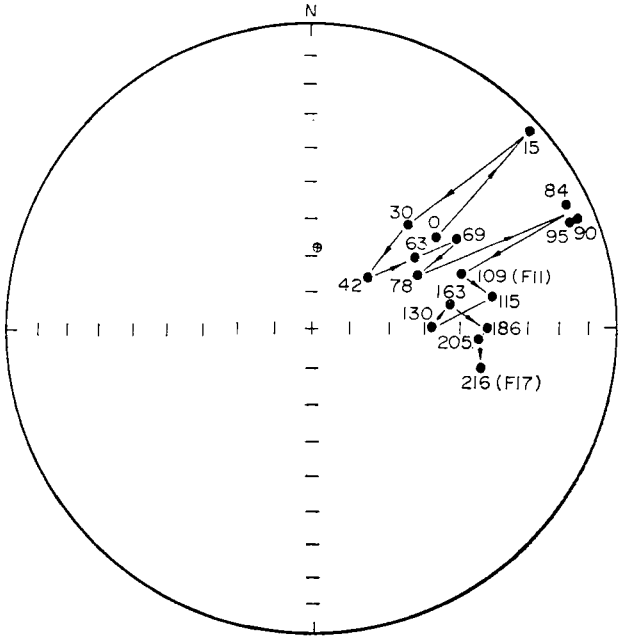


FIG. 5. The variation of the direction of magnetization in specimens from the columnar section of the fine grained sandstone at site F1. Specimens demagnetized using peak field of 250 Oe. The numbers 0 to 216 indicate the thickness of the section in centimetres from bottom to top. Specimen code numbers of the upper part of the section are shown in parentheses.

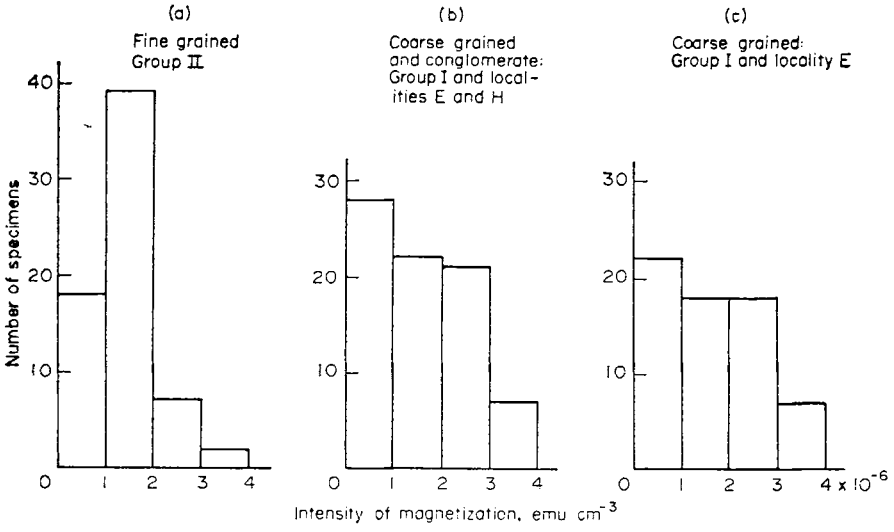


FIG. 6. Histograms of the magnitudes of NRM of two main rock types of the Lamotte formation. (*a*) Fine grained rock, (*b*) coarse grained rock including the conglomerate, (*c*) coarse grained rock without the conglomerate.

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the coarse grained rock, constructed with and without the conglomerate specimens (Fig. 6(*b*) and (*c*)), are those of a Poissonian distribution as found by applying the chi-square test in both cases. The results of this simple statistical analysis suggest that the NRMs of the two types of rock differ significantly from each other.

5. Stability and nature of the NRM

The determination of the age of the NRM is a basic requirement in palaeomagnetic study. If, in the case of sediments, a stable NRM with statistically valid direction is found, it could be the original depositional remanent magnetization (DRM), or a secondary chemical remanent magnetization (CRM) acquired either at the time of deposition and compaction of sediments or at a much later time. Alternatively it could be a viscous partial thermoremanent magnetization (VPTRM) acquired as a result of burial under subsequently eroded younger sediments.

Figs 3 and 4 show that the mean direction of magnetization of each group diverges from the direction of the Earth's present field. This and the fact that the results for specimens obtained in the second sampling confirm those for the first sampling indicates that a distinct and stable component exists in these specimens. This conclusion is also supported by the fact that in more strongly magnetized specimens there was no change in directions close to the mean direction of the groups even on application of very high peak alternating fields.

The histograms of the magnitude of the NRM of the two types of the Lamotte formation given in Fig. 6 and the corresponding statistical analysis indicate that the coarse grained rock and the conglomerate possess similar characteristics, distinct from those of the fine grained group II. This is also illustrated by the coercivity spectra shown in Fig. 7. The specimens from group II clearly contain a very stable component which remains unchanged even on application of a peak field of 2800 Oe. The magnetization of group I and locality E is unstable, disappearing in peak fields of 700 and 350 Oe respectively. The smaller stability of the 'clean' NRM of group I in comparison with that of group II was confirmed by remeasuring the NRM of representative specimens after storage in the laboratory for a period of one year.

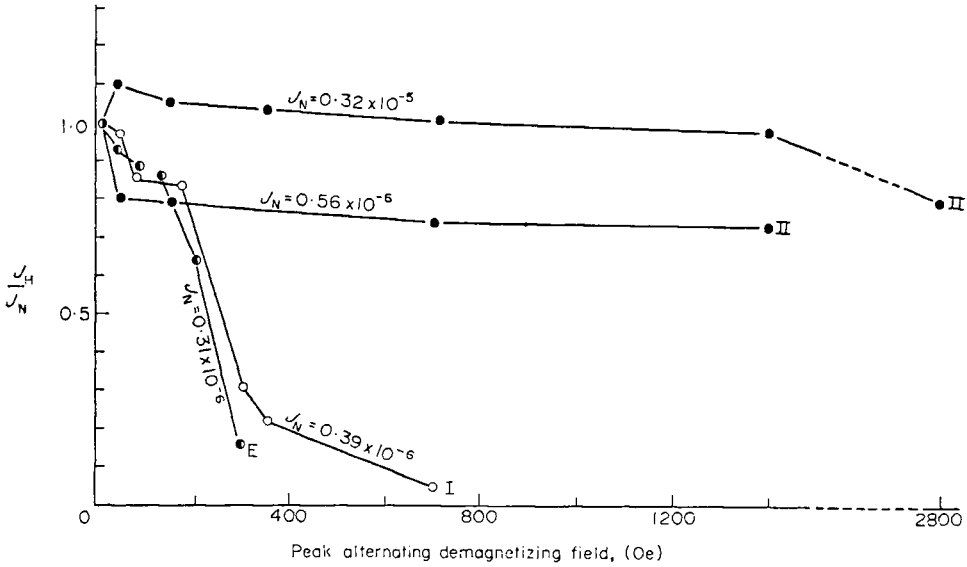


FIG. 7. Typical AF demagnetization curves. Units of NRM (J_N), emu cm^{-3}.

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The directional changes in group I were of the order of several tens of degrees while those of group II amounted only to a few degrees, well within the limits of experimental error.

These results show that there is a significant difference between the magnetization of the coarse grained and fine grained parts of the formation. A clue to this difference is obtained from the investigation of magnetic minerals in the two types of the formation, which is described in the next section.

6. Identification of magnetic minerals

In view of the very low concentration of magnetic material in the sandstone, a concentrated magnetic fraction was prepared by extracting the material from powdered rock with a suitable extractor magnet. The concentrated material was then investigated using the following methods.

6.1 *Microscopic studies*

The extracted magnetic grains were embedded in the surface of suitably prepared translucent resin cylinders of 2·5-cm diameter from which polished sections were prepared and investigated by microscopic reflection technique. Typical results are shown in Fig. 8 which shows microphotographs obtained using oil immersion at 800 magnification. It was established that despite the difference in texture of the two main types of the Lamotte sandstone (coarse versus fine), there was generally no significant difference in the size of the magnetic grains in the two types of rock. The grain diameters varied from less than 1μ to about 40μ in aggregates of smaller crystals. The predominant sizes ranged from 5 to 30μ, averaging about 15μ. The magnetic material was found to consist entirely of phases of the ilmenite–haematite series described by the formula $x\mathrm{FeTiO_3}.(1-x)\mathrm{Fe_2O_3}(0 \leqslant x \leqslant 1)$.

The presence of the following magnetic minerals was established: Haematite in the form of elongated hairlike bodies 5 to 30μ long and about 2 to 5μ thick was identified in the material from group II (Fig. 8(*a*)). Aggregates of very fine grained haematite possibly in a superparamagnetic state (sizes $\sim 0{\cdot}25\mu$ which is the limit of optical resolution) were found in all the palaeomagnetic groups (Fig. 8(*b*)). Irregular crystals of haematite (diameters $\sim 20\mu$) with small rutile inclusions and corresponding titanohaematite crystals containing exsolution lamellae of haematite occurred in both groups I and II (Fig. 8(*c*)).

Other members of the ilmenite–haematite series were present in several textural forms. One form was that of ultra-fine grains of haematite embedded in a darker ground mass of haemoilmenite. Another form consisted of rounded grains with irregular margins, but apparently of cubic crystalline form and containing dark, intermediate, and bright lamellae (Fig. 8(*d*)). This form is interpreted as being made up of three components developed by exsolution from homogeneous solid solution of ilmenite–haematite. The darker lamellae consisted probably of ilmenite or haemoilmenite ($x > 0{\cdot}5$). The lamellae with intermediate shade of colour consisted of titanohaematite with $x < 0{\cdot}5$ and a third very bright component was pure haematite. The first and second components also took the form of spindles, sometimes in a seriate arrangement and at times in complex intergrowth with each other. The titanohaematite with complex intergrowth phases seemed more common in the coarse grained rock represented by the palaeomagnetic group I than in the fine grained type of group II. On the other hand titanohaematite without complex micro-intergrowths, but with large or medium size ilmenite, haemoilmenite or rutile lamellae appeared to predominate in group II. Although, as stated above, there was little difference in the magnetic grain sizes in the two types of sandstone, a few larger grains of titanohaematite (diameters $\sim 100\mu$) were found in the material extracted from the coarse-grained rock, while very small grains of pure haematite seemed to

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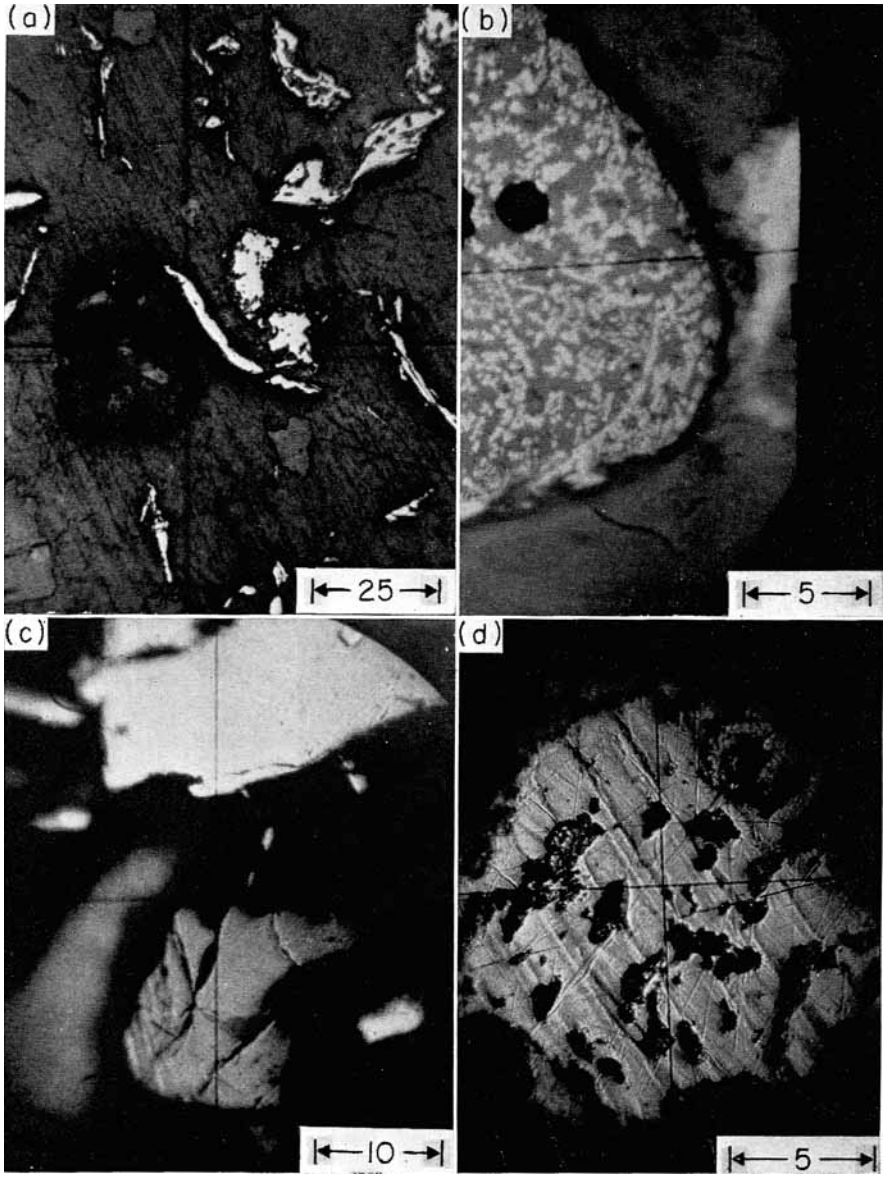


FIG. 8. Microphotographs under reflected light of polished magnetic grains extracted from virgin rock. (*a*) Haematite in the form of hairlike bodies (centre and lower half of picture), and haematite grains with rutile inclusions (upper left of picture). (*b*) Aggregates of very fine grained haematite possibly in a superparamagnetic state. (*c*) Haematite crystal (bright white) and titanohaematite (dark gray) containing exsolution lamellae of haematite (bright streaks). (*d*) A typical micro-texture resulting from unmixing of ilmenite–haematite solid solution. The black lines consist of ilmenite, the neighbouring bright lines of haematite, and those of intermediate shading of titanohaematite.

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dominate the material from group II. Unfortunately, due to paucity of magnetic material in the sandstone it was not possible to obtain repeat extracts of the material to permit a detailed comparison of the three palaeomagnetic groups. Nevertheless it was established that homogeneous titanohaematite was common in the coarse grained sandstone, while group II though revealing a predominance of haematite in in the form shown in Fig. 8(*a*), contained some titanohaematite with exsolved ilmenite or haemoilmenite lamellae. Group I and locality E, on the other hand, though containing some haematite, showed, apart from the predominance of homogeneous titanohaematite, the presence of all other ilmenite–haematite phases. No magnetite was detected in any of the material though traces of cubic crystals with optical properties similar to those of haematite were identified as an alteration product of magnetite to haematite.

The difference in the magnetic mineralogy of the two palaeomagnetic groups suggests that the two types of sandstone have passed through different processes during the past geologic ages. It would appear that the predominance of one phase of the ilmenite–haematite series or the other is a clue to understanding the origin of their magnetization.

6.2 *X-ray diffraction and emission spectrographic studies*

The X-ray diffraction studies of mixed material from groups I and II confirmed presence of haematite. Optical emission spectrographic studies confirmed the presence of iron and titanium as the only magnetic ions in the Lamotte formation.

6.3 *Curie temperatures*

Fig. 9 gives results of typical Curie point determinations. The predominance of a phase with the Curie point near 725 °C is indicated in the fine grained material, while the coarse grained material contains also a phase with the Curie temperature near 550 °C. In fact for ten specimens tested from both groups I and II, all ten contained material with the Curie temperature near 725 °C, while only three contained phases with the Curie temperature at 675 °C and six in the 550° to 600 °C temperature range. Five specimens revealed the presence of the Curie point near 250 °C and the two least frequent phases were those with Curie points near 325° and 625 °C.

There are reasons for believing that the Curie temperature of haematite is higher

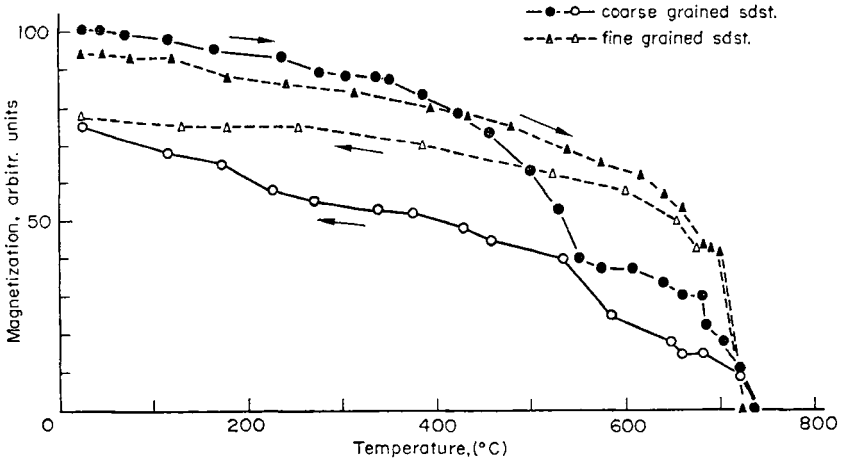


FIG. 9. Determination of Curie temperatures. Heating conducted in nominal vacuum (about 10^{-3} mm). Magnetizing field 1000 Oe. Arrows indicate the heating and cooling process.

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than 675 °C. Fig. 10 shows the results of least squares analysis of the data obtained by Nagata & Akimoto (1956) for synthetic samples of members of ilmenite–haematite series. The solid straight line represents the relation between the Curie temperatures and chemical composition of ilmenite–haematite series originally given by Nagata and Akimoto. The dashed line represents the least squares fit for the values of synthetic samples (full circles). This line yields a Curie temperature of 696 °C for $x = 0$ and gives a better approximation to the data obtained for natural minerals (open circles) than the solid line of Nagata and Akimoto. The relation shows that the observed Curie temperature at 550 °C corresponds to titanohaematite with 17 to 20 per cent ilmenite. This titanohaematite is one of the two final products of unmixing of ilmenite–haematite solid solution (Edwards 1954) and is not conspicuous in the thermomagnetic curve for the fine-grained sandstone shown in Fig. 9. The other less significant Curie temperatures are probably related to the various phases of ilmenite–haematite series according to the relation represented by the dashed line in Fig. 10. The presence of the phase with Curie point at 250 °C is unexpected in sediments and is usually associated with igneous rocks which suffered rapid chilling after a period of relatively slow cooling. The Curie temperature at 675 °C would correspond to ilmenohaematite with about 6 per cent of ilmenite according to the relation obtained. The temperatures 325° and 625 °C can be attributed either to different phases of the ilmenite–haematite series which seems unlikely, or to some process of homogenization or exsolution of the phases present.

A pertinent question is whether the Curie temperature of 725 °C obtained for haematite is not a result of contamination by iron derived from the drill barrel or from the equipment used in crushing the rock material, or is alternatively a result of incorrect calibration of the Curie balance. Actually the method used in crushing the sandstone was specifically designed to eliminate the possibility of contamination and the same method was employed in producing the material for X-ray studies which did not reveal any foreign material. The precautions taken in cleaning the cores after drilling in the field were extreme. The Curie balance was recalibrated using natural iron, magnetite and nickel and the original calibration curve was reproduced exactly.

The magnetism of haematite has been explained by Dzyaloshinski (1958) and Moriya (1960) in terms of a spin-canted moment arising from the intrinsic slight departure of the spins in the two sublattices of haematite from their antiparallel configuration. Néel (1955) suggested on the other hand a lattice defect model leading to pseudo-ferrimagnetism of haematite. Porath (1968) gave experimental evidence that remanent magnetization of haematite is, in fact, due to uncompensated antiferromagnetic sublattices and Li (1956) and Jacobs & Bean (1958) also produced arguments in favour of this model.

Smith & Fuller (1967) suggested that the magnetic properties of haematite could be explained in terms of a compromise between Néel and Dzyaloshinski–Moriya models. They produced evidence that magnetism of αFe_2O_3 is probably due to a hard structure sensitive lattice defect moment present below the true Néel point of haematite at 725 °C and also due to a softer spin-canted moment with a Curie temperature at 675 °C. The stable fraction of remanence is related to the lattice defect moment.

Recent experiments of Schmidt (1970) suggest that the low coercivity fraction of remanent magnetization in single crystals of natural haematite is oriented along the basal plane and may be associated with the spin-canted moment. On the other hand Dunlop (1970) showed that in fine grained synthetic haematites the basal plane anisotropy is annealed out at 950 °C, leading to increase in coercivity and suggesting that the defect moment is annealed out and the spin-canted (intrinsic) moment is more stable.

The present data support Néel's suggestion according to which, as a result of lattice defects, haematite behaves as a pseudo-ferrimagnetic material with a Curie

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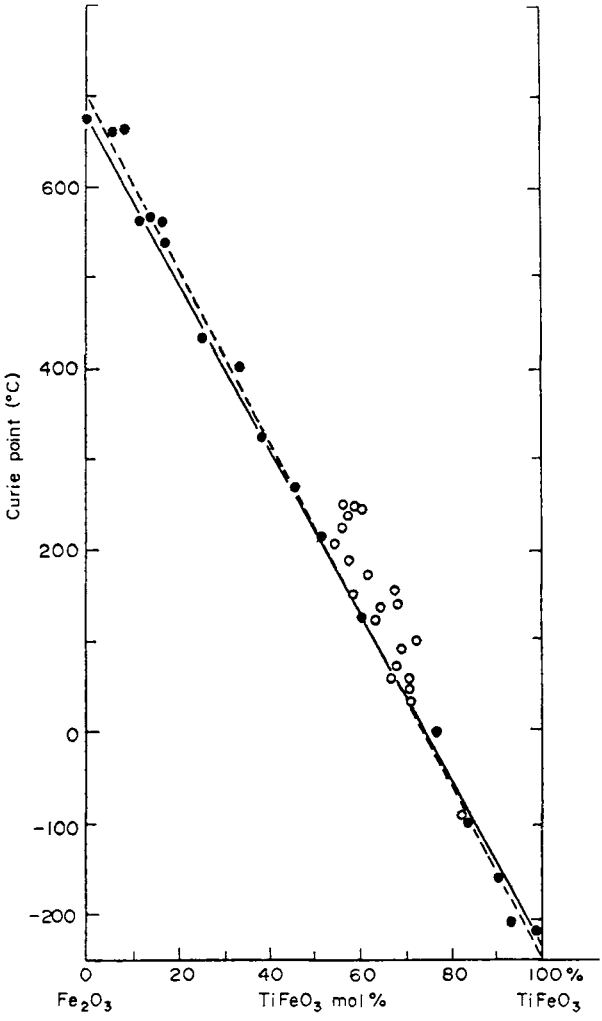


FIG. 10. Relation between Curie temperature and chemical composition of ilmenite–haematite. Solid line and the data indicated by full and open circles after Nagata & Akimoto (1956). Dashed line: least squares fit for synthetic samples shown by full circles. Natural minerals are shown by open circles.

temperature coinciding with its Néel point at 725 °C. The work of Uyeda (1958) and subsequently of Hedley (1968) supports this suggestion.

6.4 *Other magnetic properties*

The presence of haematite and of ilmenite–haematite solid solution is suggested by the observed magnitude of saturation magnetization amounting to $7{\cdot}6 \pm 5{\cdot}2$ emu g^{-1}. The high standard error arises from the fact that the Curie balance used in obtaining this result was designed for Curie point determinations only and could not easily be adapted for measurement of absolute values of saturation magnetization. The result obtained is, of course, too high to be that of haematite ($\sim 0{\cdot}5$ emu g^{-1}), but in view of the error involved may well represent haematite or, for the higher limit, an ilmenite–haematite solid solution (Nagata 1961). In any case, this value is far removed from the values associated with magnetite, titanomagnetite,

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or maghaemite. A further confirmation of absence of these minerals was obtained from low temperature analysis which yielded the Morin transition temperature near or somewhat below $-24\,^{\circ}C$ which corresponds to presence of haematite with some titanium cations in its structure.

An average weak field susceptibility value of 5×10^{-6} emu cm^{-3} was obtained from susceptibility measurements of several cylindrical cores, conducted on a commercial susceptibility bridge. This value is not far from those obtained for weakly magnetized sediments which contain haematite or titanohaematite.

Finally, it must be mentioned that the extracted magnetic material, when examined in ordinary light, possessed a high reflectivity and an intense silvery lustre, suggesting that αFe_2O_3 occurs predominantly in the form of specular haematite.

7. Thermal demagnetization of NRM

It is commonly known that when dealing with sediments thermal cleaning is preferable to AF demagnetization. If, after its deposition, the Lamotte formation had been subjected to prolonged reheating, a secondary magnetization would have been produced which might be resistant to AF treatment and could only be removed by thermal demagnetization. The observed NRM might in such a case be in part a VPTRM produced as a result of burial of the formation and subsequent cooling due to uplift and erosion. An accumulation of Palaeozoic and Mesozoic sediments above the Lamotte sandstone is estimated not to have exceeded 3 km (Ojakangas 1960) and would be expected to have gradually increased the temperature of the formation to a maximum of about 120 °C. On theoretical grounds (Néel 1949; Stacey 1963) and subsequent experimental evidence (Chamalaun 1964; Briden 1965) the probable duration ($\sim$100 My) of burial of the formation at this depth would be equivalent to subjecting it to a temperature of about 470 °C when considered in terms of duration of laboratory experiments. Thus to remove the VPTRM, it would be necessary to carry out thermal cleaning by heating the sandstone specimens to this temperature. Unfortunately, in view of the low magnitude of the NRM of the formation, such a procedure was practicable on a very limited scale and only seven more strongly magnetized specimens were heated. Four of the specimens, each two representing groups I and II, were heated to 130 °C and revealed no change in direction. An increase in intensity of about 30 per cent was observed in the specimens of group I, probably caused by removal of a partially self-reversed component and confirming the presence of a haemoilmenite phase with a low blocking temperature (Vincenz 1968). The remaining three specimens representing groups I and II and locality E were taken up to 470 °C with an intermediate step at 378 °C which was repeated to check consistency. The intensities and their directions were measured using a PAR spinner magnetometer and it was found that heating to 470 °C reduced the intensities to approximately one-third of their virgin values. When a spinner magnetometer is used the observations give four values of each of the three components of intensity X, Y and Z and these are averaged to obtain the mean values of the components. In the present case the magnitudes of X, Y and Z observed after heating revealed very high dispersions. Since for each of the four consecutive sets of values of X, Y and Z, the corresponding D and I can be computed, it is possible to consider the observations as four independent directions and to treat them by standard statistical techniques for dispersion on a unit sphere. The results of this analysis are summarized in Table 1. The Table gives mean declinations and inclinations obtained after each heating from four observations of X, Y and Z components. The accuracy of observations is given in terms of the semi-angle of confidence, α_{95} (Fisher 1953) and the corresponding RMS angular deviation of the mean, ε (Wilson 1969). Evidently, in view of the low intensities involved, little or no significance can be attached to the

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Table 1

Results of thermal demagnetization

T	Group I				Locality E				Group II			
	D	I	α_{95}	ε	D	I	α_{95}	ε	D	I	α_{95}	ε
25	335·5	32·0	11·0	4·1	134·0	71·0	7·0	2·6	75·2	0	9·0	3·5
378(1)	321·4	8·8	9·5	3·6	160·0	29·5	52·5	17·7	83·9	14·1	26·3	9·7
378(2)	314·7	18·8	27·8	10·2	139·5	15·0	21·4	8·0	97·5	17·0	28·0	10·3
470	333·0	16·1	69·3	21·7	80·8	10·2	41·4	14·6	130·7	46·5	71·8	22·5

Notes:

T = temperature in °C; (1) and (2) refer to the first and second heating respectively;

D = Declination of the mean direction of magnetization derived from four sets of observations;

I = Corresponding inclination.

α_{95} = Corresponding semiangle of cone of 95 per cent confidence (Fisher 1953).

ε = RMS angular deviation of the mean (Wilson 1959).

mean directions obtained after heating to 470 °C. Moreover, on applying the randomness test (Watson 1956; Vincenz & Bruckshaw 1960) to four directions obtained for each specimen at 470 °C, the unit vector sum obtained in each case was 2·905, 3·491 and 2·856 for group I, locality E and group II respectively. Two of these values are exceeded by the corresponding randomness limit at the 5 per cent probability level ($R_0 = 3{\cdot}10$). Thus the directions of magnetization of the specimens from groups I and II at 470° are not significant and have to be interpreted as the consequence of random fluctuations in the low accuracy observations. In the case of group II the deviations of the four observed directions from their mean allow for the probable value of the direction to be within both subgroups of group II. Although no element of randomness is associated with the mean direction at 470 °C of the specimen from locality E, both α_{95} and ε are large and thus little significance can be attached to this result.

It is concluded that thermal cleaning could not reliably be applied to the very low NRM of the Lamotte sandstone. The fact that AF demagnetization has yielded significant mean directions suggests that the effect of VPTRM was probably insignificant or, if present, was to a large extent erased by AF treatment. Specifically, the results suggest that the fine grained rock carries very little or no VPTRM. The magnetization of this part of the Lamotte formation is thus either a DRM or a CRM. In view of the smaller stability of the coarse grained part of the formation one would expect it to carry some VPTRM. The behaviour of the specimen from locality E gives some indication that this locality may have preserved a small amount of magnetization with a direction associated with group II. The behaviour of group I suggests, however, that this part of the formation must have been completely remagnetized, though the result at 470 °C is inconclusive.

8. The conglomerate test

The conglomerate test was applied to 13 samples of the lowest part of the formation. The directions of magnetization are shown in Fig. 11. Although some of the virgin directions are close to the present day field direction, others resemble those of the virgin NRM of group I (compare with Fig. 4) and the mean direction for 13 samples (indicated by a triangle) is different from the direction of the present day field. On AF demagnetization with a peak field of 250 Oe the directions of specimens H411 and H312 have moved towards the 'clean' mean direction of group I. The direction of specimen H211 is relatively close to the mean for group I and did not change

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significantly on heating to 130 °C. The mean direction for the three demagnetized specimens falls close to the 'clean' mean of group I.

The randomness test (Watson 1956; Vincenz & Bruckshaw 1960) was applied separately to the mean of 13 virgin directions and to the mean direction of three demagnetized samples. The values of the unit vector sum R obtained were 11·4 and 2·97 respectively, both exceeding the corresponding values of R_0 (see Table 3) at the 1 per cent probability level. This suggests that the magnetization of the conglomerate is not random in direction.

Since the mechanical forces involved in the formation of the conglomerate are very large and far exceed the magnetic forces acting during the deposition, they should always destroy any DRM in the conglomerate, the magnetization of which should be random. In the present case, since the direction of magnetization is not random, the magnetization must have been acquired after deposition. The similarity of the directions of group I and of the conglomerate (compare Figs 4 and 11) suggests that they contain comparable magnetic constituents, which is consistent with the common origin of both units. Since the conglomerate underlies the beds represented by specimens of group I, these beds must have been remagnetized jointly with the conglomerate after deposition. Thus the main component of NRM of group I is of secondary origin and does not represent the original DRM.

9. Origin of the natural remanent magnetization

The fact that the NRM of the Lamotte formation is weak unfortunately limits the scope of its experimental investigation. The dilemma of weak magnetization may in part be avoided by imposing remanent magnetization in the laboratory and studying its behaviour. Thus the nature of magnetization of groups I, II, and E was

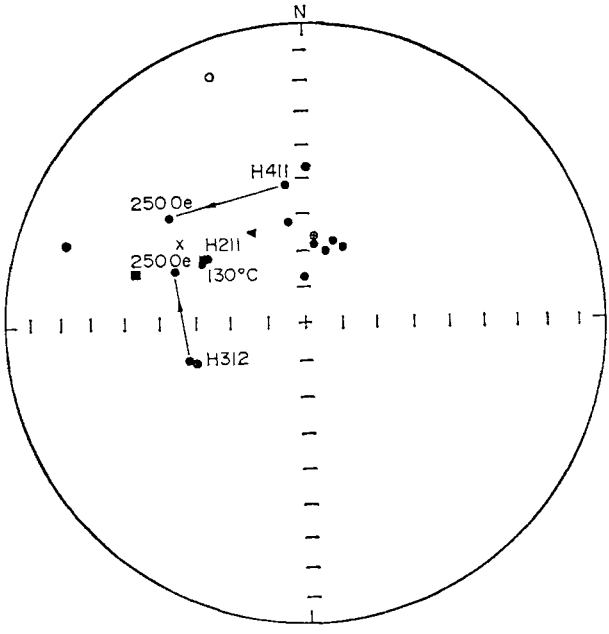


FIG. 11. Conglomerate test using directions of 13 specimens. Specimens H411 and H312 were demagnetized in a peak field of 250 Oe. Specimen H211 was thermally demagnetized in 130 °C. Cross, mean direction for specimens H411, H312, and H211; triangle, mean direction for 13 virgin specimens; square, mean direction of group I.

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investigated by imposing isothermal remanent magnetization (IRM) and thermo-remanent magnetization (TRM) on representative rock specimens from the three groups and studying their behaviour on AF and thermal demagnetization.

9.1 *Behaviour of IRM*

The procedure was to demagnetize completely the relevant specimens by AF demagnetization and to apply a steady field in successive steps up to the maximum value of 10·5 kOe, measuring the IRM produced after each step. This was done for virgin rock as well as for rock heated in the laboratory at 735 °C for the total time of 6 hr and cooled in zero field. Three specimens from group I, three from group II and one from locality E were used in this experiment and the results are shown in Fig. 12. Since there was little difference in the behaviour of each of the three individual specimens in groups I and II respectively, the three magnetization curves for each group were normalized with respect to the average IRM acquired in 10·5 kOe and then were averaged. The curve for the heated rock gives the result for six specimens representing both groups I and II.

Clearly groups I and E behave very similarly and heated rock gives an exaggerated

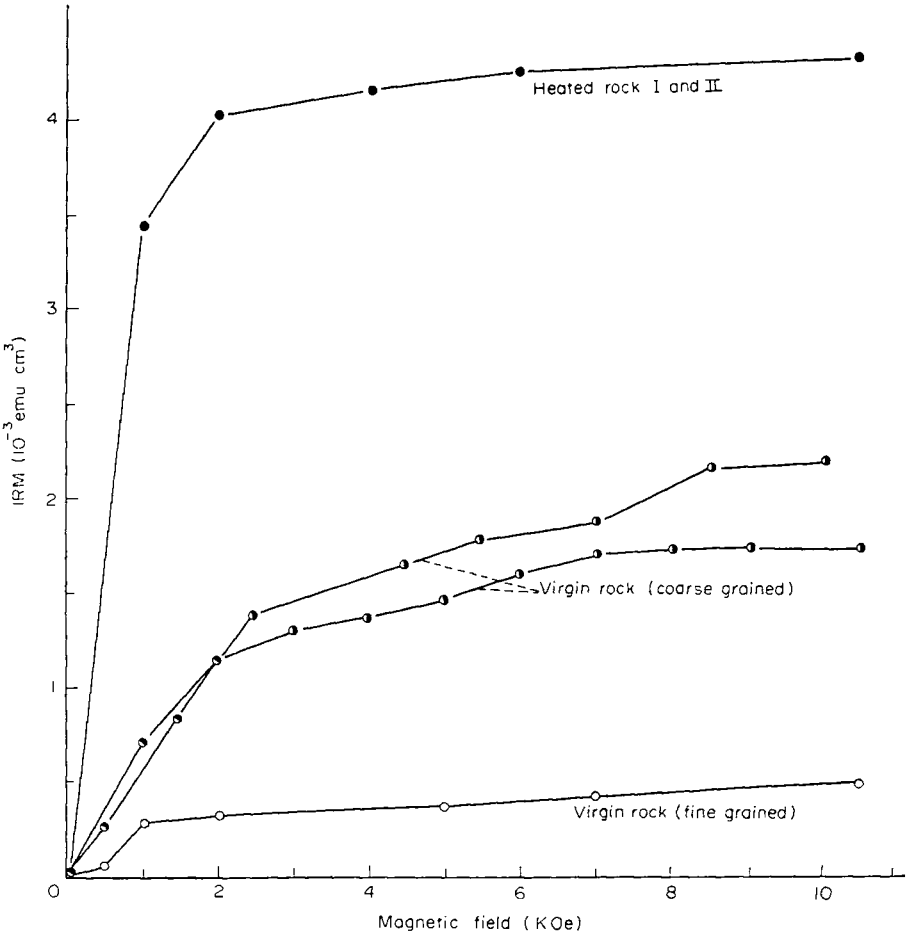


FIG. 12. Acquisition of IRM by both virgin and heated rock. Curves for groups I and II are average curves for three specimens, normalized to the IRM acquired in 10·5 kOe.

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effect of the two, but the fine grained rock has a very distinct behaviour, acquiring a much weaker IRM. The results show that the main increase in IRM of the virgin specimens takes place in fields below 2 kOe. This rapid initial increase is unexpected in rocks containing haematite or ilmenohaematite (Roquet 1954). It suggests the presence of magnetic constituents with coercivities lower than those of the main constituents of the Lamotte formation. On the other hand none of the curves become horizontal, showing that saturation has not been achieved, which is typical for haematite and titanohaematite. However, heat treatment has evidently produced material of lower coercivity, capable of carrying an IRM a few times larger than that acquired before heating.

Fig. 13 shows the results of AF demagnetization of the IRM produced in 10·5 kOe. Again normalized average curves are shown for both groups I and II and for heated rock. Clearly, the IRM of groups I and E is much less stable than that of group II and the magnetic constituents of heated rock have a low coercivity about the same as that of group I. Since the behaviour of rock from locality E is very similar to that of specimens from group I it is legitimate to conclude that it has comparable magnetic constituents. Consequently specimens from locality E will not be included in subsequent experiments in which they will be assumed to be represented by the behaviour of specimens from group I.

Comparison of the result shown in Fig. 12 with the histograms of Fig. 6 indicates that the IRM of group II is about 100 times greater than its NRM and the IRM of group I 1000 times greater than its NRM. Evidently the small saturation magnetization of haematite ($\sim 0{\cdot}5$ emu g^{-1}) cannot account for this tremendous difference. An IRM much greater than the NRM would be expected in a rock which has a high concentration of magnetic material, but the Lamotte formation possesses a very low content of magnetic material. Akimoto (1954) showed that saturation magnetization of haemoilmenite and to a lesser extent of ilmenohaematite is much greater that that

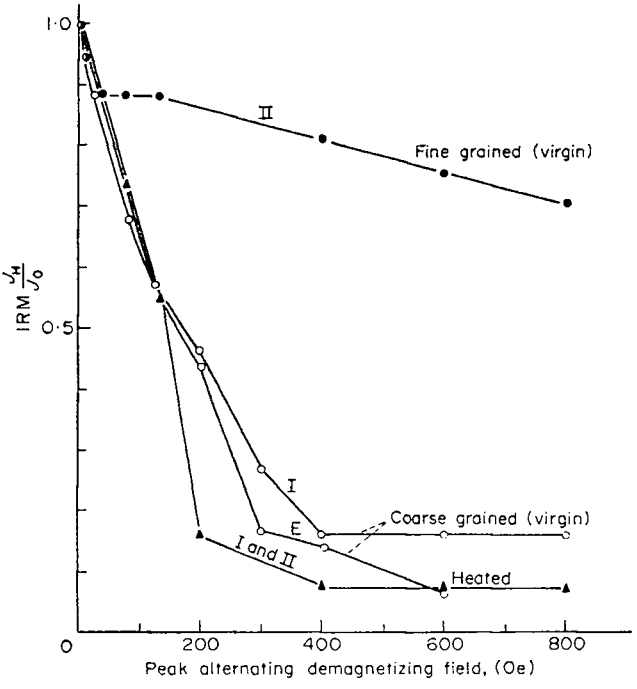


Fig. 13. AF demagnetization of IRM of specimens described in Fig. 12. Ordinates normalized to IRM at zero peak field, J_0.

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of haematite. Ilmenohaematite constitutes the main magnetic mineral of group I and this could partly account for the higher IRM. However, the rapid initial increase of IRM observed is not characteristic of the ilmenite–haematite series. We conclude, therefore, that the rock must contain a low coercivity material in a demagnetized state, which carries no NRM, but becomes strongly magnetized by the isothermal process. This conclusion is supported by polished section studies which revealed presence of material in a very finely divided state (Fig. 8(*b*)). In addition the unexpectedly strong IRM may also have been acquired by the fine haematite and titanohaematite lamellae and microintergrowths in the magnetic grains of group I. These may also be in part in a demagnetized state in the virgin rock because their size is too small to have acquired any CRM by the nucleation process (Nagata 1961). It would appear thus that the more intense IRM acquired by the specimens of group I is mainly accounted for by the presence of homogeneous ilmenohaematite (Curie temperature about 550 °C) and titanohaematite with microintergrowths.

The stabilities of IRM in groups I and II were compared by investigating the decay of IRM over a period of 8 days. Table 2 shows the results of this investigation. It is evident that IRM of group I has decayed faster than that of group II, and this suggests that group I must contain more material in a demagnetized state than group II.

The difference in stability of groups I and II can be seen by comparing the AF demagnetization spectra of NRM and IRM of the two groups as shown in Figs 7 and 13. The similarity of demagnetization curves of NRM and IRM for each group is striking.

Fig. 14 shows results of thermal demagnetization of IRM acquired in 10·5 kOe. Here again, normalized averages for three specimens from each group as well as for six specimens after thermal treatment at 735 °C are shown. The results show that group I and heated rock possess a smaller stability than group II. The curve for group II reveals, however, presence of an unstable component with a very low blocking temperature, as suggested by the initial rapid decay of magnetization of the NRM of group II. Hence this confirms the presence of superparamagnetic material also in group II, since the low blocking temperature is related to grain size which must be near its critical value (Stephenson 1967). Nevertheless, the overall stability of group I is much smaller and this is most likely related to the presence of homogeneous titanohaematite (Curie temperature 550 °C) and of the other ilmenite–haematite phases.

The most important results are those obtained for the heated specimens. The thermal demagnetization curves of thermally treated specimens from both groups and of virgin specimens from group I resemble each other and both reveal a blocking temperature at about 450 °C. In every case heat treatment has brought the material of group II into a state similar to that of group I, but did not change the properties of group I. As will be described below, the cause of this change is probably related to the production and homogenization of ilmenohaematite.

Table 2

Decay of IRM with time

Time (days)	IRM (10^{-3} emu cm^{-3})		IRM_t/IRM_o	
0	1·78	0·730	1·00	1·00
1	1·44	0·672	0·81	0·92
8	1·37	0·650	0·77	0·89

Notes:

IRM_o (emu cm^{-3}) = intensity of IRM measured immediately after acquisition in a field of 10·5 kOe.

IRM_t (emu cm^{-3}) = intensity of IRM measured after time t.

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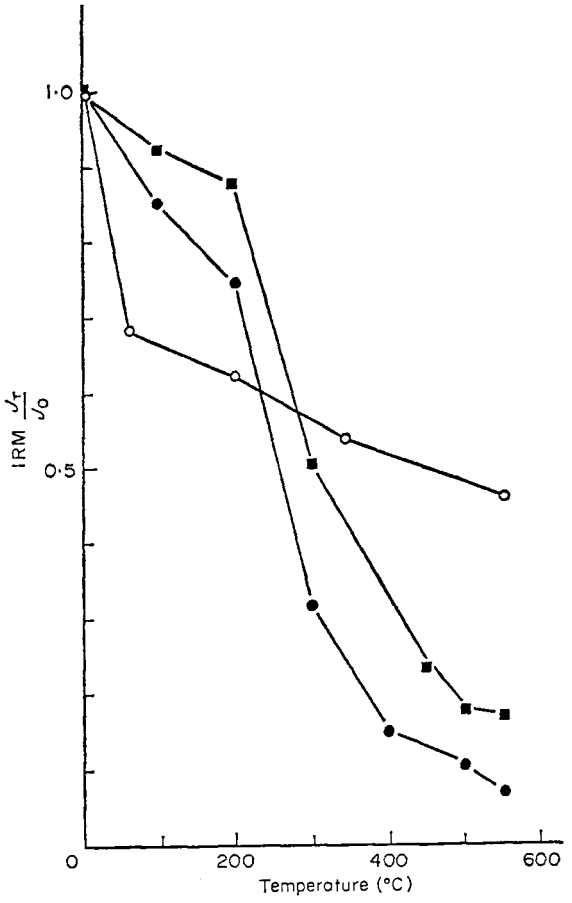


FIG. 14. Thermal demagnetization of IRM acquired in 10·5 kOe. Ordinates normalized to IRM at room temperature, J_0. Full circles, virgin rock of group I; open circles, virgin rock of group II; squares, heated rock of groups I and II.

9.2 *Behaviour of TRM*

The specimens used in the study of IRM were subjected to thermoremanent magnetization in a field of 0·38 Oe. The TRM produced was measured and demagnetized stepwise by both AF and thermal treatment and the results are shown in Figs 15 and 16. Evidently, as a result of heating, groups I and II behave similarly, confirming the deduction that heat treatment has changed group II and made it behave as group I. The small dissimilarity between the two curves in both Figs 15 and 16 can be accounted for by insufficient heat treatment which lasted only for 6 hr plus the time taken for imposing TRM. This explains probably the somewhat higher blocking temperature of TRM of heated specimens of group II than that of specimens of group I.

9.3 *Discussion*

The fact that heat treatment has brought the magnetic material of group II into a state similar to that of group I requires special consideration. This is apparent in the increase of IRM produced after heating, in the decrease of stability of heated specimens of group II as shown on AF and thermal demagnetization of IRM, as well as in the similarity of TRM imposed on heat treated specimens of both groups.

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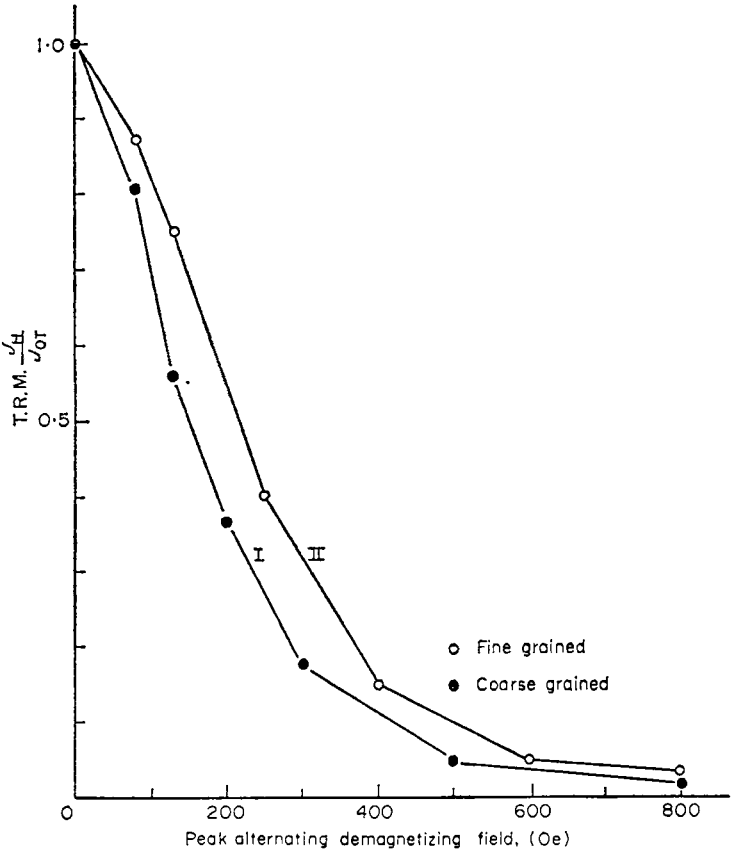


FIG. 15. AF demagnetization of TRM acquired in a field of 0·38 Oe. Ordinates normalized to TRM at zero peak magnetizing field, J_{OT}.

The cause of this behaviour can be explained by one or all of three possible mechanisms. The first possibility is that heat treatment may have produced some magnetite by reduction of haematite. This is an unlikely explanation, because no change in weak field susceptibility was observed after heating. The second alternative is that of growth of superparamagnetic grains to a size exceeding the critical volume. This hypothesis might explain the increase of IRM and the low coercivity of heated specimens of group II, but it offers no explanation for the thermal demagnetization curves of IRM and TRM which show that new material, which can carry strong magnetization, has been produced, with blocking temperature at about 450 °C. The third alternative relates to production and homogenization of ilmenohaematite. It was shown that heat treatment caused the specimens of group II to behave in a manner similar to the virgin specimens of group I, but did not significantly change the properties of specimens of group I. Since virgin specimens of group I contain the ilmenohaematite phase of the ilmenite–haematite series, heat treatment apparently created such a phase in specimens of group II and did not bring any change in the magnetic minerals of group I. It would appear that this last alternative represents the most probable mechanism for the behaviour of heated specimens of both groups.

The process can be explained in terms of the solvus curve for the ilmenite–haematite solid solution $x\mathrm{FeTiO_3}.(1-x)\mathrm{Fe_2O_3}(\mathrm{O} \leqslant x \leqslant 1)$ proposed by Uyeda (1958). According to this curve an ilmenohaematite phase with 15 to 20 per cent ilmenite in

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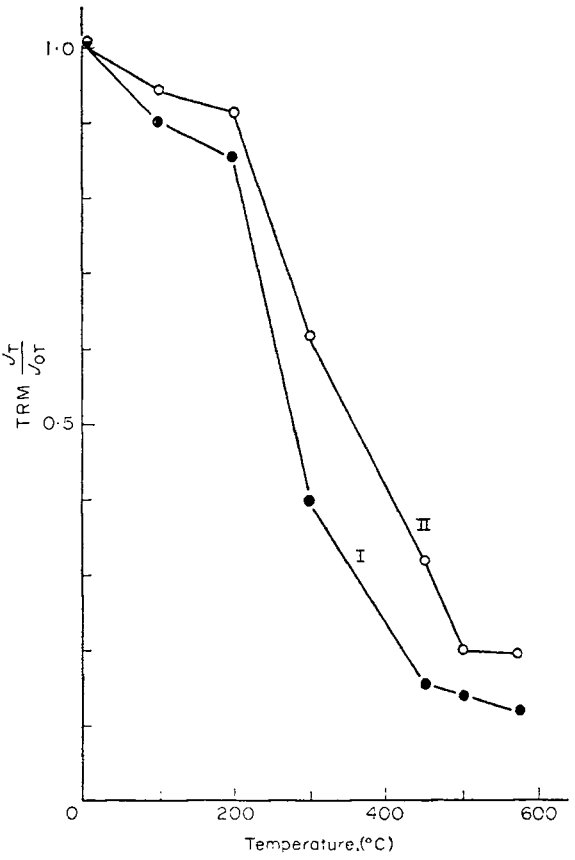


FIG. 16. Thermal demagnetization of TRM acquired in a field of 0·38 Oe. Ordinates normalized to TRM at room temperature, J_{OT}.

haematite can be maintained between room temperature and 735 °C. This situation corresponds to the behaviour of group I in which the original ilmenohaematite phase with about 20 per cent ilmenite has not been changed by heat treatment. The behaviour of group II on baking at 735 °C corresponds to the situation in which an homogeneous ilmenohaematite phase with 15 to 20 per cent ilmenite has been created by the process of solution of ilmenite lamellae into haematite or titanohaematite, leading to an increase in saturation magnetization and the corresponding observed large IRM. In fact, within limits of experimental error, the observed blocking temperatures of heated specimens correspond roughly with the Curie temperature of 550 °C for homogeneous titanohaematite with about 20 per cent ilmenite (Fig. 10). Insufficient heat treatment would lead to incomplete re-solution of ilmenite into ilmenohaematite and thus to a little higher blocking temperature as shown in Fig. 16.

A conclusions of prime importance is that a process which takes place in nature has been simulated on a laboratory scale and heat treatment of a relatively porous sediment has produced a situation which in natural conditions would favour creation of CRM.

It is concluded thus that the magnetization of the coarse grained group I and locality E is mainly associated with homogeneous ilmenohaematite and is largely a CRM of secondary origin. On the other hand the magnetization of the fine grained group II is mainly associated with haematite and represents the original DRM. This evidence, combined with the fact that the mean directions of magnetization of the

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three groups are significantly different from each other, suggests that the NRM of each group was acquired at a different time. Moreover the fact that the intensities of the coarse grained rock follow Poissonian distribution and those of the fine grained logarithmic normal suggests a difference in the deposition mode and magnetization process of the two groups. Actually, even though ilmenohaematite with ex-solution lamellae is more common in the fine grained rock, it is also present in the coarse grained sandstone. In point of fact magnetic grains in both types of sandstone were derived from igneous rocks and must have originally been oriented by the Cambrian geomagnetic field. For some reason, however, at a later date chemical processes caused re-solution of the lamellae in the coarse grained rock only, producing homogeneous titanohaematite and the resulting CRM imposed by the ambient field.

Table 3

Parameters of group mean directions of magnetizations

Name	N	D	I	R	R_0	k	α_{95}	H
3(a) Results of observations—first sampling								
Group I	11	322·5	49·1	9·1	5·03	5·4	21·7	130
Group II	12	41·3	28·7	9·5	5·28	10·8	11·6	250
Locality E	7	145·6	41·0	6·6	4·18	10·6	15·4	88
3(b) Results of observations—second sampling								
*Group I	24	291·0	41·1	22·5	7·86	10·5	7·8	250
*Group IIa†	33	52·2	26·7	30·3	9·23	10·2	7·6	250
Group IIa‡	45	54·3	29·5	41·1	10·76	11·3	6·6	250
*Group IIb	9	95·9	45·0	8·8	4·76	30·3	9·1	250
Group IIb^{ss}	14	96·0	46·3	13·7	5·98	31·0	6·4	250
*IIa+IIb	2$^{=}$	71·4	37·9	1·9		8·6	7·47	250
*Locality E	4	140·6	14·3	3·9	3·1	20·3	19·5	130
Locality H	13	331·6	62·3	11·4	(6·84)	7·5	16·2	—
Locality H	3	303·5	50·2	2·97	(2·84)	80·2	13·7	250 (T=130°C for H211)
3(c) Results of combined directions from Table 3(a) and 3(b)								
Group I	34	295·3	44·3	26·6	9·37	4·5	13·1	130 & 250
Locality E	11	142·9	31·4	10·2	5·28	10·4	12·7	88 & 130
Group II	42	59·9	31·9	37·5	10·40	9·0	7·7	250

Notes:

N = Total number of specimens used in calculating the mean, except when groups IIa and IIb are combined and the computation is carried out at group level.

D = Declination of the mean direction of magnetization.

I = Inclination of the mean direction of magnetization.

R = Vector sum of N unit vectors.

R_0 = A value of R which is exceeded by observed R at 5 per cent probability level if the directions of N specimens in a population are random (Watson 1956; Vincenz & Bruckshaw 1960). (1 per cent probability level used for locality H).

k = $(N-1)/(N-R)$ = Estimate of the precision parameter (Fisher 1953).

α_{95} = Semiangle of cone of 95 per cent confidence for mean direction (Fisher 1953).

H = Peak alternating demagnetizing field in oersted.

* = Directions used in calculating palaeomagnetic pole positions

† Includes the 12 specimens from Table 3(a).

‡ Includes 12 specimens from columnar section.

ss Includes five specimens from columnar section.

= Number of groups, each group given unit weight.

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10. Analysis of directions

10.1 *Statistical tests*

Table 3(*a*), (*b*) and (*c*) gives the parameters of the mean direction of magnetization for each group, including the two stages of sampling. The results for group II, given in Table 3(*b*), are calculated at group level using the data for subgroups II(a) and II(b), representing two stratigraphically distinct parts of the formation. The 17 specimens obtained in sampling the columnar section at locality F were not used in the calculations of the above mean directions. This is because these 17 specimens represent only one site (F1) at locality F. On the other hand, the specimens from site F1, obtained prior to sampling the columnar section, have been included in these calculations. The results for the conglomerate at locality H are given separately, both for 13 virgin samples and for three demagnetized samples. The estimate of the precision parameter, k, and the angle of the cone of confidence, α_{95}, were calculated on the assumption that the directions used in the analysis obey Fisher's distribution (Fisher 1953). Fig. 17 summarizes all the mean directions of the data obtained in the two stages of sampling.

10.2 *Palaeomagnetic deductions*

Fig. 18 shows the positions of computed palaeomagnetic poles with their 95 per cent confidence ovals. Pole IIa is evidently an Upper Cambrian pole corresponding to the lower columnar section of the fine grained part of the Lamotte sandstone. At a later date, corresponding to the upper columnar section, the pole moved westward to the position IIb. The locality E carrying secondary demagnetization represents a Carboniferous pole. Group I, representing the majority of coarse grained specimens, was shown to carry a CRM, but the palaeomagnetic significance of its direction of magnetization is not clear. In view of its generally low coercivity it was more difficult

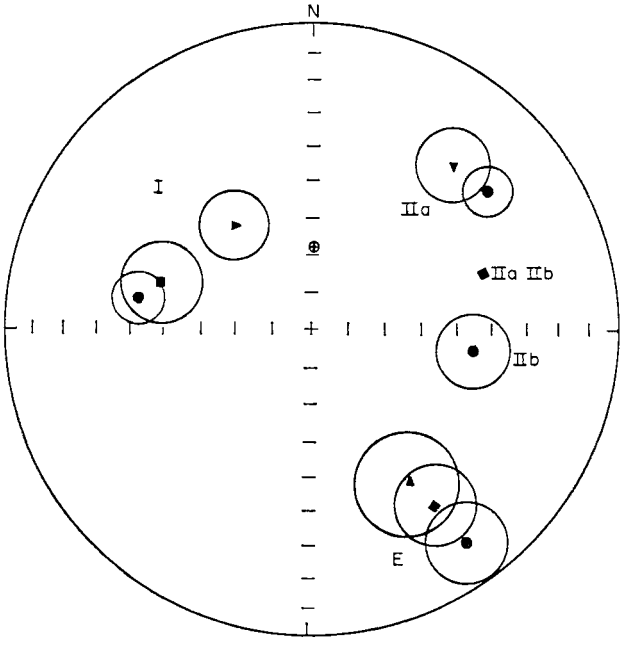


FIG. 17. Group mean magnetization directions with their respective 95 per cent confidence circles. Triangles: first sampling. Full circles: second sampling. Squares: first and second samplings combined. The direction of the present geomagnetic field is given by a crossed open circle.

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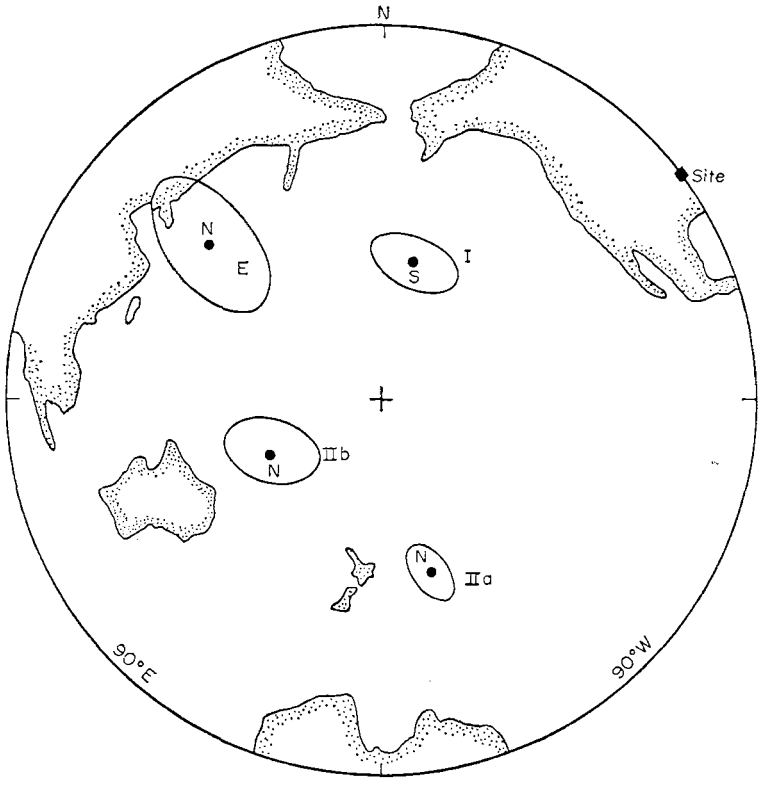


FIG. 18. Palaeomagnetic pole positions calculated from mean directions of magnetization given in Table 3.

to demagnetize it and still retain some measurable magnetization. Possibly the peak fields used did not completely remove recent secondary components. It could represent a Precambrian pole, but this is unlikely. More probably, this direction represents a composite effect of self-reversed Cambrian magnetization and of a later larger Carboniferous or possibly Triassic component. Unfortunately, no data are available to support this suggestion, though presence of complex microintergrowths of titanohaematite and the partial self-reversal effect observed on heating to 130 °C do make this idea attractive.

The earlier described result of thermal cleaning of one specimen from locality E suggested that Carboniferous remagnetization of locality E was possibly only partial, resulting from burial under younger sediments. Had thermal cleaning of all Lamotte specimens been at all practicable, the Carboniferous magnetization of locality E, if incomplete, would have been erased, revealing the original magnetization represented by group II. The result obtained with one specimen supports thus the conclusion that the direction of group II is that of the Upper Cambrian geomagnetic field.

The final palaeomagnetic conclusions of this investigation are summarized in Table 4 and Fig. 19 which give a comparison of the Lamotte palaeomagnetic poles with the other poles for Cambrian rocks of North America. The continuous curve (1) represents Collinson & Runcorn's (1960) polar-wandering path for North America, while the dotted curve gives the more detailed path due to Irving (1964). Curve (2) represents the approximate polar-wandering path for Europe as given by Collinson and Runcorn.

The Lamotte poles IIa, IIb and E clearly lie on the North American polar-wandering

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Table 4

Palaeomagnetic pole positions

Worker	Rock unit	Palaeomagnetic pole	Polarity
Day & Runcorn (1955)	Tapeats Sandstone	22° S, 153° W	+
Howell & Martinez (1957)	Wilberns Formation	00°, 150° W	?
Howell & Martinez (1957)	Sawatch Sandy Dolomite	47° N, 125° E(2°, 4°)	+
Collinson & Runcorn (1960)	Lodore Formation	23° S, 154° W (4°, 7°)	M
Collinson & Runcorn (1960)	Deadwood Formation	47° N, 117° E (4°, 7°)	+
Yaskawa, Vincenz & Guja (1966)	Wichita Mountains	2° N, 146° W	M
Ku *et al.* (1967)	Wichita Mountains	33·5° N, 134° E (7°, 15°)	+
Spall (1968)	Wichita Mountains	34° N, 118° E (4°, 7°)	M
Spall (1968)	Wichita Mountains	2° N, 147° E (6°, 9°)	M
Al-Khafaji & Vincenz (this paper)	Lamotte Formation Group II*a*	38° S, 168° W (4°, 8°)	+
Al-Khafaji & Vincenz (this paper)	Lamotte Formation Group II*b*	11·5° S, 115° E (7°, 11°)	+
Al-Khafaji & Vincenz (this paper)	Lamotte Formation Group I	30° N, 173° W (8°, 9°)	−
Al-Khafaji & Vincenz (this paper)	Lamotte Formation Locality E	32° N, 137° E (10°, 20°)	+

Note:

The figures in parentheses in column 3 give the semi-axes of the elliptical error at a probability of 95 per cent whenever available. M denotes mixed polarity.

path which might, however, require a slight modification. Comparison of the average longitudes of the pole positions obtained by Collinson and Runcorn from European rocks with those calculated from the Lamotte formation indicates a difference of +13° between the Carboniferous rocks in Europe and locality E and +5° and +18° for the Cambrian rocks using the data obtained from subgroups IIa and IIb respectively. This gives an average difference of +12° in longitude between the European pole positions and those obtained from the Lamotte formation. Runcorn (1956, 1965) postulated a relative westward drift of North America of about 24° with respect to Europe since the Mesozoic. Although the difference of 12° includes the somewhat scanty data from locality E, the positive sign of the angle confirms Runcorn's hypothesis of the drifting apart of the two continents since at least the early Mesozoic and evidently even since the Carboniferous. The fact that the Lamotte poles obtained from groups IIa and IIb differ in longitude from the European Cambrian poles might suggest that North America had separated from Laurasia already in the Upper Cambrian. The Soviet data for Eastern USSR (see Creer 1968), however, do not support this suggestion. They give an average Cambrian pole just south of Australia and other lower Palaeozoic poles inside Australia (Fig. 19). The mean Siberian pole is just east of New Zealand, fairly close to the pole for group IIa, while the mean pole for European USSR is close to the pole for group I. Creer (1968) showed that the difference between the Lower Palaeozoic poles for Europe and Western USSR and those for Eastern USSR can be accounted for by the widespread remagnetization of the former during Upper Palaeozoic. He suggested that this remagnetization affected also the North American Lower Palaeozoic rocks. The fact that only two Lamotte poles can justifiably be regarded as representing the Cambrian geomagnetic fields supports Creer's suggestion. In the light of the Lamotte data five of the nine previously reported poles shown in Table 4 and Fig. 19 appear to be Cambrian, while the remaining four are Upper Palaeozoic or even Mesozoic. The Lower Palaeozoic poles for Eastern USSR are, however, situated west of the pole obtained from group IIa. This suggests that, contrary to Creer's view, it seems after all necessary to invoke the tectonic event by which the two originally separate parts of USSR became welded together along the

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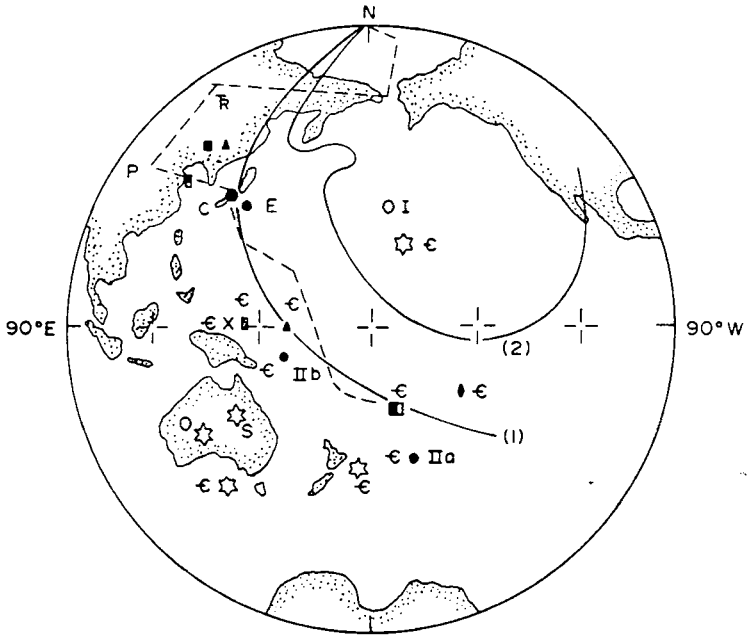


FIG. 19. Comparison of pole positions obtained from the Lamotte formation and from other Palaeozoic rocks of North America and USSR. The curves indicate the polar-wandering paths for North America (1) and Europe (2). Continuous curves after Collinson & Runcorn (1960), dotted curve after Irving (1964).

line of Urals, but that this event must have taken place before the separation of North America from Laurasia. Conclusions of this nature are of course highly speculative, but it seems indisputable that a considerable part of both Eurasian and North American Lower Palaeozoic rocks became remagnetized during Upper Palaeozoic. According to Vincenz & Yaskawa (1972) the greater part of Wichita Mountains granites have lost their original magnetization carried by primary titanomagnetite which has been oxidized to titanohaematite carrying a secondary CRM. The Lamotte data support Creer's suggestion that the remagnetization was facilitated by the low geographic latitudes of the affected parts of Laurasia, resulting in wide laterization processes and formation of red sediments; after all the coarse grained part of the Lamotte sandstone is reddish. It would seem that, where for some reason red sediments were not formed, the chemical remagnetization process did not take place or was incomplete. To test their validity these suggestions require closer study. A call for such investigations has already been made by Creer (1969). In the case of the Lamotte formation there is certainly evidence that its fine grained *white* part appears to carry the original DRM, while its coarse grained *reddish* part carries a CRM and has been remagnetized. Moreover there is geological evidence (Ojakangas 1960) that the coarse grained rock has originated directly from the local pre-existing

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Precambrian rocks, while the fine grained sandstone is *not* of local origin and was deposited by off-shore currents paralleling the shoreline of the Lamotte sea.

11. Conclusions

The central facts derived from the investigation of the Lamotte formation are:

1. The formation is a very weakly magnetized predominantly white sandstone.

2. The coarse grained somewhat reddish lower part of the formation carries secondary magnetization of chemical origin, associated mainly with ilmenohaematite.

3. One locality in the coarse grained part reflects the direction of the Carboniferous geomagnetic field.

4. The fine grained white upper part of the formation carries the original DRM associated mainly with specular haematite and some ilmenohaematite with ilmenite lamellae and reflects the direction of the Upper Cambrian geomagnetic field.

5. The results suggest that stable remanence in natural haematite is associated with a lattice defect moment with a Curie temperature of 725 °C.

6. The fine grained sandstone subdivides into two stratigraphically differentiated parts, defining two distinct phases of the Upper Cambrian field.

7. The pole positions determined from one locality in the coarse grained part and from the two parts of the fine grained sandstone fit the polar-wandering path for North American rocks.

8. Comparison of Lamotte palaeomagnetic poles with those for Europe suggests that North America separated from Laurasia already in Upper Palaeozoic.

9. Non-coincidence of the Cambrian Lamotte poles with the Cambrian poles for Eastern USSR suggests that the welding together of two parts of Eurasia took place before the separation of North America from the Laurasian supercontinent.

10. The results suggest that red Lower Palaeozoic sediments of North America carry magnetization of mainly secondary origin, but some white sediments seem to have retained the original DRM.

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Appendix

Sampling localities

Where possible, grid references on the 15-min 1 : 62,500 topographic sheets of Missouri are given, the quadrangle name appearing in parenthesis. Generally, one to two cylindrical specimens (2·49 cm in diameter and 2·28 cm long) were obtained from the oriented diamond drilled cores.

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Locality C: Four sites along and near highway '00' near Mine Lamotte about 6 km north of Fredericktown; 11 cores with one to five cores from each site; white fine grained sandstone exposed in creek cuts and on shores of man-made lakes; sections 28, 29, 30 and 33, T.33 N., R.7 E. (Fredericktown).

Locality D: Five sites along highway 32 about 13 km ENE from Farmington; 18 cores with three to five cores per site; coarse grained arkose sandstone; sections 24 and 25, T.35 N., R.7 E. (Farmington).

Locality E: Four sites; nine cores with one to three cores per site; a 10 metres thick section exposed in the road cut of highway '00' near Mine Lamotte; lithology as in D; section 29, T.33 N., R.7 E. (Fredericktown).

Locality B: Two sites with two cores each in road cut about 5·5 km east of intersection of highway K and highway 72; coarse grained arkose sandstone; section 4, T.33 N., R.6 E. (Fredericktown).

Locality F: Eight sites with a total of 28 cores, three to five cores per site and a columnar section of 17 cores at F1 from an extensive exposure along Cedar Creek where it intersects highway 32 about 11 km west of Bismarck; white fine grained sandstone; section 7, T.35 N., R.3 E. (Bonne Terre).

Locality G: Four sites, with a total of 10 cores, two to three cores per site; exposure in a small stream crossing highway 21 in Caledonia, about 14 km west of Bismarck; very coarse grained sandstone, darker in colour than sandstone at locality E. (Potosi).

Locality H: Five sites with a total of 10 cores, one to two cores per site; exposure close to locality G; conglomerate of the same colour as the sandstone at locality *G.* (Potosi).

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